# LahjaMT at AlexandriaX-2026: A Context-Aware English-to-Dialectal Arabic Machine Translation System with Lightweight Routing of LoRA Experts

**Mohamed A. Abdallah***
Newgiza University
mohamedabdallah9850@gmail.com

**Samhaa R. El-Beltagy**
Nawy and Newgiza University
samhaa@computer.org

## Abstract

Dialectal Arabic dialogue translation requires modeling both regional variation and conversational context under uneven supervision. We introduce LahjaMT, a compact system for translating English dialogue into thirteen country-level Arabic varieties. LahjaMT adapts the 3.09B-parameter NileChat-3B-Base model with LoRA, conditions translation on dialogue history and metadata, and routes each dialect to a checkpoint–prompt expert selected on the held-out selection set. The constrained system shares one backbone across all varieties; the unconstrained system changes only the Libyan and Sudanese routes, replacing them with specialists trained on auxiliary data. LahjaMT ranked second in the constrained track with 28.30 spBLEU and third in the unconstrained track with 28.54 spBLEU.

## 1 Introduction

Across centuries, the Arab world has developed a rich and geographically diverse collection of Arabic varieties, shaped by distinct histories, cultures, and patterns of social interaction. Arabic, therefore, cannot be treated as a single homogeneous translation target. Modeling this lexical, morphosyntactic, pragmatic, and orthographic variation remains challenging, particularly because dialectal Arabic–English parallel resources are unevenly distributed across varieties (Sajjad et al., 2020).

Dialogue introduces a second axis of variation: the appropriate rendering of a short turn may depend on the preceding conversation, the topic under discussion, and the identities of the speaker and addressee.

AlexandriaX-2026 Subtask 1 brings these challenges together, where the task is to translate selected English dialogue turns into one of thirteen country-level Arabic varieties spanning eleven conversational domains (El Mekki et al., 2026a). Each source turn is accompanied by conversational and sociolinguistic context. The task further tests broader dialectal coverage through Libyan and Sudanese Arabic, for which no training conversations are provided. It includes a constrained track, limited to the officially released data and models of at most 5B parameters, and an unconstrained track in which both restrictions are lifted.

We approached the task with an economical goal: to determine how far a compact backbone could be pushed through careful input design, adaptation, and model selection. LahjaMT adapts the 3B-parameter NileChat-3B-Base with LoRA, conditions translation on conversational context and metadata, and routes each dialect to a checkpoint–prompt expert selected on held-out data. At inference, all routes use beam search with a beam size of four, while the thirteen experts share one backbone and six lightweight adapters.

Despite its compact scale, LahjaMT ranked second in the constrained track and third in the unconstrained track. Separately from the official shared-task evaluation, our experiments showed that it outperformed configurations based on the much larger gpt-oss-20b and gpt-oss-120b models, underscoring the value of task-specific adaptation over scale alone. Beyond the ranked submissions, this paper documents the expert construction and promotion process, the targeted use of auxiliary data for Libyan and Sudanese Arabic, and quantified negative results for candidate selection and post-editing.[1]

## 2 Task, Data and Metrics

**Task.** Given a selected English dialogue turn, its available preceding English context, a target coun-

*First and corresponding author.

[1]Our code is available at https://github.com/m-abdallah98/LahjaMT.



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try code, and associated metadata, the system must generate a translation in the corresponding dialectal Arabic variety. The task covers thirteen varieties: EG, JO, LB, LY, MA, MR, OM, PS, SA, SD, SY, TN, and YE. Systems are ranked by spBLEU (Goyal et al., 2022), with chrF++ (Popović, 2017) also reported; country-level scores are averaged equally to obtain the overall score (El Mekki et al., 2026a).

**Official data.** Alexandria (El Mekki et al., 2026b) consists of multi-turn conversations, which we convert into turn-level examples while retaining conversation identifiers. Initial LoRA training uses the 66,480-turn training split, covering eleven varieties as Libyan (LY) and Sudanese (SD) have no training or development conversations. For continuation training after the development phase, we combine the 12,250-turn development split with 8,670 turns, which are 60% of the labeled public-test split (which contains 14,442 turns). We keep nearly 40% for held-out selection, yielding a 5,772-turn selection set. This set is excluded from gradient training and the in-context demonstration pool, and is used exclusively for checkpoint, prompt-configuration, and routing decisions. Final evaluation uses the separate private test set, containing 14,459 turns across 4,673 conversations.

## 3 LahjaMT System

Figure 1 summarizes the offline expert construction and routing, while Figure 2 shows how a single expert is selected and applied at inference time.

### 3.1 Backbone and Adaptation

We use NileChat-3B-Base (El Mekki et al., 2025), which is a decoder-only model obtained by continuing the pretraining of Qwen2.5-3B (Qwen et al., 2025). NileChat further adapts Qwen2.5-3B using data centered on Egyptian and Moroccan Arabic.

We adapt the frozen backbone with LoRA (Hu et al., 2022). For a selected projection $W$ of shape $d_{\text{out}} \times d_{\text{in}}$, LoRA learns two low-rank matrices: $A$ of shape $r \times d_{\text{in}}$, which maps the input into an $r$-dimensional bottleneck, and $B$ of shape $d_{\text{out}} \times r$, which maps it back to the original output dimension. The effective weight is therefore $W' = W + (\alpha/r)\, BA$, where $\alpha/r$ is a scaling factor that controls the adapter's contribution relative to the frozen projection.

During training, only the adapter matrices $A$ and $B$ are updated while $W$ remains frozen. Although LoRA can target different subsets of model modules, our preliminary experiments obtained the strongest held-out performance with $r = 16$, $\alpha = 32$, dropout of 0.05, and adaptation of all seven projection modules in every Transformer block. This joint adaptation of the attention and feed-forward layers outperformed attention-only targeting. Adaptation adds approximately 29.9 million trainable parameters, 0.96% of the resulting 3.12B-parameter model, and produces an approximately 114 MB adapter, making it economical to retain multiple checkpoints as experts.

### 3.2 Context-Aware Input Representation

To expose the task structure directly to the adapted backbone, we format each instance as a single instruction block containing four components: two English–Arabic demonstrations; metadata specifying the country, dialect, domain, speaker, speaker–addressee gender direction, and persona or participant roles when enabled; up to three preceding English turns with speaker labels; and the current English turn. Seven explicit instructions constrain the output, including preserving meaning and named entities, preferring the requested local dialect over MSA, and returning only the translation. Training loss is applied only to the target response tokens. Appendix A reproduces the complete template.

Together with the LoRA checkpoint, the prompt configuration defines an expert. We compare three configurations: P1 selects two demonstrations deterministically using a seeded hash of the example identifier, restricted to the same country and domain, matching the format used during fine-tuning; P2 replaces them with two semantically retrieved demonstrations (Agrawal et al., 2023); and P3 combines retrieval with persona and participant-role metadata. For P2 and P3, all-MiniLM-L6-v2 (Reimers and Gurevych, 2019; Wang et al., 2020) encodes the English source turns. Candidates are restricted to the same country and domain, backing off to the country when necessary, and ranked by cosine similarity with small bonuses for matching the sub-dialect, gender direction, and speaker role. Demonstrations from distinct conversations are preferred when possible. The pool comprises the official training and development splits and the 60% continuation-training portion of the labeled public-test split; all held-out conversations are excluded.

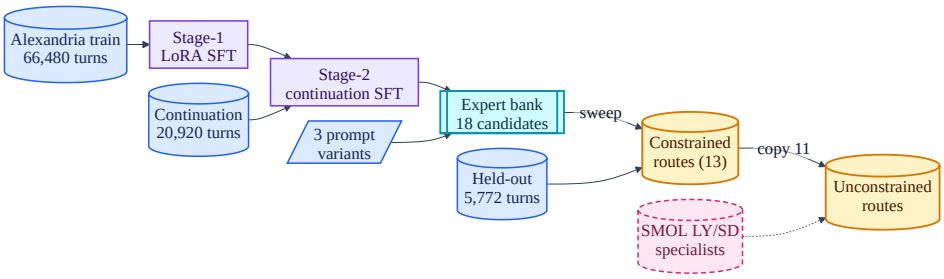



Figure 1: LahjaMT expert construction. Two-stage LoRA training produces six checkpoints, which are paired with three prompt configurations to form 18 candidates and routed by dialect using held-out data. The dashed SMOL branch is used only for the unconstrained submission. Edge labels denote the per-dialect selection (*sweep*) and the copying of the eleven shared routes (*copy 11*) to the unconstrained table.

### 3.3 Two-Stage Training

**Stage 1.** We fine-tune on the 66,480 training turns for three epochs using a learning rate of $2 \times 10^{-4}$, cosine scheduling, a warm-up ratio of 0.03, weight decay of 0.01, an effective batch size of 8, a maximum sequence length of 2,048, and BF16 precision. We save checkpoints every 100 optimization steps and evaluate them on the development split using the official metrics. The checkpoint that achieves the strongest performance becomes the parent adapter for continuation training.

**Stage 2.** Starting from the Stage-1 parent, we continue training for three epochs on 20,920 turns: the 12,250-turn development split and the 8,670-turn continuation portion of the labeled public-test split. We reduce the learning rate to $2 \times 10^{-5}$ while keeping all other settings unchanged. Six checkpoints are retained for expert construction. The best single Stage-2 expert improves held-out spBLEU by 2.91 over the Stage-1 parent; dialect routing contributes a further 0.24 (Section 4.2). All runs use BF16 on one NVIDIA RTX 5090.

### 3.4 Expert Bank and Dialect Routing

Stage 2 begins from the best-performing Stage-1 parent adapter. We evaluate checkpoints throughout the continuation run and select the checkpoint with the highest macro-averaged spBLEU as the default. However, performance varies across dialects, and no single checkpoint is optimal for all of them. To capture this variation, we retain six checkpoints and combine them with three prompt configurations, yielding an expert bank of 18 checkpoint–prompt pairs. Dialect-specific evaluation further shows that the optimal checkpoint differs across SA, MR, EG, and TN.

We construct a routing table on the held-out selection set using a conservative selection procedure. For each dialect, we first select the best prompt configuration at the default checkpoint. We assign a dialect to an alternative checkpoint only when its best checkpoint–prompt pair improves on the corresponding default pair by at least 0.15 spBLEU points. Under this criterion, seven dialects retain the default checkpoint, while six (OM, YE, LY, MR, SA, and TN) are assigned to alternative checkpoints. Notably, a dialect may achieve its highest raw score at another checkpoint yet remain at the default if the improvement falls below the promotion threshold. We adopt this conservative threshold because each dialect-specific held-out subset contains only approximately 444 turns, making small score differences insufficient evidence for changing the route. Appendix B provides the complete routing map.

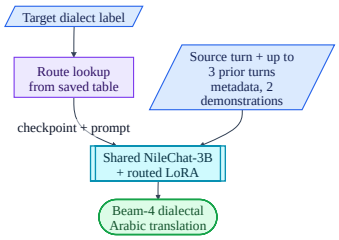



Figure 2: LahjaMT inference.

At inference, the provided dialect label deterministically retrieves the corresponding checkpoint–prompt pair. All routes use beam search with four beams, a length penalty of 1.0, a repetition penalty of 1.05, and up to 120 new tokens. No Arabic normalization is applied; post-processing only strips template markers and removes predominantly Latin-script lines.

| Track | Rank | spBLEU | chrF++ |
|---|---|---|---|
| Constrained | 2 | 28.30 | 43.81 |
| Unconstrained | 3 | 28.54 | 44.02 |

Table 1: Official private test results for LahjaMT, macro-averaged across all thirteen dialects.

### 3.5 Unconstrained Specialists

The unconstrained system modifies only two of the thirteen constrained routes. For LY and SD, we extract English-original pairs from the SmolSent and SmolDoc components of SMOL (Caswell et al., 2025), using the `ayl` and `apd` language subsets, respectively. After applying pre-processing, this yields 2,469 LY and 2,461 SD pairs.

To limit forgetting, we mix each specialist set with three copies of the corresponding dialect's in-domain examples, adding 1,995 replayed turns for LY and 1,992 for SD. Starting from the constrained LoRA checkpoint, we train the specialists for two epochs at a learning rate of $5 \times 10^{-6}$. A specialist replaces its constrained route only if it achieves a held-out spBLEU gain of at least 0.30 while maintaining a chrF++ change of at least $-0.10$. Both specialists satisfy these criteria; the remaining eleven routes are exactly those of the constrained submission.

## 4 Results and Analysis

### 4.1 Official Results

Table 1 reports LahjaMT's official private-test results. The system ranks second in the constrained track and third in the unconstrained track. Replacing only the LY and SD routes yields gains of 0.24 spBLEU and 0.21 chrF++ over the constrained system. The test-set spBLEU gain exceeds the 0.13 improvement estimated on the held-out selection set.

### 4.2 Component Contributions

Table 2 reports internal component comparisons on the 5,772-turn held-out selection set, rather than official private-test scores. The Stage-1 routing table was selected on the development split, which covers eleven varieties, where it improved spBLEU by 0.23 over the parent adapter. We then applied this routing table unchanged to the separate held-out set, which covers all thirteen varieties. It routed unseen dialects through geographic proxies: LY through TN and SD through EG. This routed system scored 27.00 spBLEU, slightly below the parent adapter's 27.04, indicating that the development gain did not persist. Continuation training provides the largest improvement, increasing held-out spBLEU by 2.91 over the Stage-1 parent. Reconstructing the dialect routes after continuation contributes a further 0.24 without additional training, while the LY and SD specialists contribute another 0.13. These differences are calculated from unrounded scores.

| Configuration | spBLEU | chrF++ |
|---|---|---|
| Stage-1 parent adapter (step 16,600) | 27.04 | 42.90 |
| Stage-1 routed system | 27.00 | 42.85 |
| Stage-2 best single expert | 29.94 | 44.63 |
| Stage-2 + constrained routing | 30.18 | 44.81 |
| + LY/SD specialists (unconstr.) | 30.32 | 44.90 |

Table 2: Internal component analysis on the 5,772-turn held-out selection set, macro-averaged across all dialects.

Prompt configuration has less influence than checkpoint selection. On the Stage-1 development split, restoring the metadata block improves spBLEU by 0.39, while replacing fixed demonstrations with semantically retrieved examples contributes 0.13. Adding persona information produces no measurable improvement. Although the aggregate scores of the three retained configurations span only 0.24 spBLEU, each performs best for at least one dialect: P1 for eight dialects, P2 for two, and P3 for three. This complementarity makes prompt-level routing beneficial.

### 4.3 Negative Results and the Routing Ceiling

A reference-dependent per-sentence oracle reveals substantial routing headroom of 36.01 spBLEU versus 30.93 for the deployed router, and none of the selectors realized it. External data helped only for LY and SD, while scaling in parameters wasn't better. Appendix C reports the full analysis, and unsuccessful alternatives in Table 4.

## 5 Conclusion

LahjaMT combines context-aware LoRA adaptation with deterministic dialect routing over checkpoint–prompt experts on a shared 3.09B-parameter backbone. It ranked second on the constrained track and third on the unconstrained track. Overall, the results favor careful task-specific adaptation and selection over model scale alone.

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## References

Ahmed Abdelali, Hamdy Mubarak, Younes Samih, Sabit Hassan, and Kareem Darwish. 2021. QADI: Arabic dialect identification in the wild. In *Proceedings of the Sixth Arabic Natural Language Processing Workshop*, pages 1–10, Kyiv, Ukraine (Virtual). Association for Computational Linguistics.

Sweta Agrawal, Chunting Zhou, Mike Lewis, Luke Zettlemoyer, and Marjan Ghazvininejad. 2023. In-context examples selection for machine translation. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 8857–8873, Toronto, Canada. Association for Computational Linguistics.

Rania Al-Sabbagh. 2024. ArzEn-MultiGenre: An aligned parallel dataset of Egyptian Arabic song lyrics, novels, and subtitles, with English translations. *Data in Brief*, 54:110271.

M. Saiful Bari et al. 2024. ALLaM: Large language models for Arabic and English. *arXiv preprint arXiv:2407.15390*.

Isaac Caswell et al. 2025. SMOL: Professionally translated parallel data for 115 under-represented languages. In *Proceedings of the Tenth Conference on Machine Translation*, pages 1103–1123, Suzhou, China. Association for Computational Linguistics.

Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8440–8451, Online. Association for Computational Linguistics.

Bryan Eikema and Wilker Aziz. 2020. Is MAP decoding all you need? the inadequacy of the mode in neural machine translation. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 4506–4520, Barcelona, Spain (Online). International Committee on Computational Linguistics.

Abdellah El Mekki, Houdaifa Atou, Omer Nacar, Shady Shehata, and Muhammad Abdul-Mageed. 2025. NileChat: Towards linguistically diverse and culturally aware LLMs for local communities. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 10967–10991, Suzhou, China. Association for Computational Linguistics.

Abdellah El Mekki, AbdelRahim A. Elmadany, Samar M. Magdy, Saad Ezzini, Mo El-Haj, Mustafa Jarrar, Zaid Alyafeai, Bernard Ghanem, and Muhammad Abdul-Mageed. 2026a. AlexandriaX 2026: The First Shared Task on Dialectal Arabic Machine Translation. In *Proceedings of The Fourth Arabic Natural Language Processing Conference: Shared Tasks*, Budapest, Hungary. Association for Computational Linguistics.

Abdellah El Mekki, Samar M. Magdy, Houdaifa Atou, Ruwa AbuHweidi, Baraah Qawasmeh, Omer Nacar, Thikra Al-hibiri, Razan Saadie, Hamzah A. Alsayadi, Nadia Ghezaiel Hammouda, Alshima Mohammed Alkhazimi, Aya Hamod, Al-Yas Yaqoob Al-Ghafri, Wesam El-Sayed, Asila Ismail al Sharji, Mohamad Ballout, Anas Belfathi, Karim Ghaddar, Serry Sibaee, and 28 others. 2026b. Alexandria: A multi-domain dialectal Arabic machine translation dataset for culturally inclusive and linguistically diverse LLMs. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 32567–32592, San Diego, California, United States. Association for Computational Linguistics.

Naman Goyal, Cynthia Gao, Vishrav Chaudhary, Peng-Jen Chen, Guillaume Wenzek, Da Ju, Sanjana Krishnan, Marc'Aurelio Ranzato, Francisco Guzmán, and Angela Fan. 2022. The FLORES-101 evaluation benchmark for low-resource and multilingual machine translation. In *Transactions of the Association for Computational Linguistics*, volume 10, pages 522–538.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.

NLLB Team. 2022. No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672*.

OpenAI. 2025. gpt-oss-120b & gpt-oss-20b model card.

Maja Popović. 2017. chrF++: words helping character n-grams. In *Proceedings of the Second Conference on Machine Translation*, pages 612–618, Copenhagen, Denmark. Association for Computational Linguistics.

Qwen et al. 2025. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pages 3982–3992, Hong Kong, China. Association for Computational Linguistics.

Hassan Sajjad, Ahmed Abdelali, Nadir Durrani, and Fahim Dalvi. 2020. AraBench: Benchmarking dialectal Arabic-English machine translation. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 5094–5107, Barcelona, Spain (Online). International Committee on Computational Linguistics.

Wenhui Wang, Furu Wei, Li Dong, Hangbo Bao, Nan Yang, and Ming Zhou. 2020. MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. In *Advances in Neural Information Processing Systems*, volume 33, pages 5776–5788.

Mitchell Wortsman, Gabriel Ilharco, Samir Ya Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S. Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, and Ludwig Schmidt. 2022. Model soups: Averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 23965–23998. PMLR.

## Appendix A Prompt Template

Examples are serialised as `### System: / ### Instruction: / ### Arabic translation:` blocks. The instruction block is:

```
Task:
Translate the current English dialogue turn
into the target dialectal Arabic variety.

Few-shot training examples:
{two demonstrations}

Metadata:
Country/config: {code}
Target dialect: {dialect}
Domain: {domain}
[Persona/Roles: {participants}]
Current speaker: {speaker}
Speaker-to-addressee gender dir.: {dir}

Previous English dialogue context:
{up to 3 numbered prior turns}

Current English turn:
{source}

Rules:
- Preserve the meaning exactly.
- Use the target local dialect, not Modern
  Standard Arabic unless natural in context.
- Follow the dialect/style pattern shown in
  the few-shot examples when relevant.
- Do not copy the few-shot examples.
- Preserve names, numbers, named entities,
  and technical terms when appropriate.
- Keep the tone appropriate for the speaker
  and domain.
- Return only the Arabic translation.
```

Each demonstration is rendered as `Example i (config=…, dialect=…, domain=…)` followed by its English and Arabic sides. The Persona/Roles line appears only in configuration P3. Demonstrations are capped at 450 characters for source and target combined.

## Appendix B Final Routing Map

Adapters are continuation checkpoints; the two specialist rows apply only to the unconstrained submission.

| Variety | Route | spBLEU | chrF++ |
|---|---|---|---|
| EG | 5,200 + P3 | 31.88 | 45.60 |
| JO | 5,200 + P1 | 35.50 | 49.12 |
| LB | 5,200 + P1 | 32.29 | 46.00 |
| LY | 6,400 + P3 | 23.39 | 39.01 |
| LY, unconstr. | + SMOL spec. | 23.86 | 39.14 |
| MA | 5,200 + P2 | 23.31 | 39.77 |
| MR | 2,000 + P1 | 17.94 | 34.34 |
| OM | 4,900 + P1 | 32.57 | 47.11 |
| PS | 5,200 + P1 | 34.26 | 48.26 |
| SA | 1,600 + P2 | 35.25 | 49.78 |
| SD | 5,200 + P1 | 26.15 | 40.97 |
| SD, unconstr. | + SMOL spec. | 27.43 | 42.06 |
| SY | 5,200 + P3 | 39.38 | 53.19 |
| TN | 7,600 + P1 | 35.23 | 47.61 |
| YE | 4,900 + P1 | 25.23 | 41.71 |
| Macro (constrained) | | 30.18 | 44.81 |
| Macro (unconstrained) | | 30.32 | 44.90 |

Table 3: Final per-dialect routing map with held-out spBLEU and chrF++.

## Appendix C Negative Results and the Routing Ceiling

The most interesting finding is that a reference-dependent per-sentence oracle reveals substantial routing headroom. For each development turn, it selects the highest-scoring output among eight Stage-1 experts by comparing every candidate with the reference translation. It obtains 36.01 spBLEU, compared with 30.93 for the deployed dialect router, a diagnostic gap of 5.08 spBLEU. Both figures are computed on the development split over this Stage-1 pool, so the analysis isolates per-sentence selection headroom; it is a diagnostic and is separate from the deployed Stage-2 system (30.18 spBLEU on the held-out set, Table 2). Because references are unavailable at inference, this oracle is non-deployable and is not reported as a system result; our highest development-phase leaderboard entry used this oracle.

The remaining headroom lies below the country level: an oracle restricted to selecting one expert per country exactly matches the deployed router,

whereas allowing reference-based selection over finer metadata groups provides a further $+0.91$ spBLEU. Moreover, the expert with the lowest aggregate score is the per-sentence oracle's most frequent choice, winning on 30% of development turns. Aggregate expert quality is therefore a poor indicator of which expert will perform best on an individual instance.

Despite this headroom, none of our deployable selectors improved performance. We initialized an XLM-R-large cross-encoder (Conneau et al., 2020) from a dialect classifier trained on the same data, fitted it to reference-derived utilities using a pairwise ranking loss, and evaluated it out of fold over conversation-disjoint partitions. It changed 48% of the routed decisions but reduced performance for all eleven development varieties. Minimum Bayes risk selection over three continuation experts (Eikema and Aziz, 2020), post-editing with an ALLaM-7B adapter trained on our generated drafts (Bari et al., 2024), and LoRA checkpoint interpolation in weight space (Wortsman et al., 2022) also reduced performance. Out-of-fold metadata routing scored $-0.17$ spBLEU relative to its in-sample counterpart, suggesting that part of the routing gain reflects selection noise. Overall, expert-level quality differences are real, but they were not predictable from the source-side, metadata, and candidate-surface features we tested.

External data was similarly ineffective outside LY and SD. An Egyptian specialist trained on 12,000 filtered subtitle pairs from ArzEn-MultiGenre (Al-Sabbagh, 2024) regressed at all nine evaluated checkpoints. A Yemeni specialist trained on dialect-identification data (Abdelali et al., 2021), back-translated with NLLB-200 (NLLB Team, 2022), also regressed. The same held-out promotion rule therefore admitted the beneficial LY and SD specialists while rejecting the harmful EG and YE alternatives. Held-out scores span more than 20 spBLEU across varieties: SY, JO, SA, and TN score highest, whereas MR, MA, and LY score lowest.

Increasing model scale without task-specific adaptation did not close the gap in our experiments. On the development split, our inference-only gpt-oss-20b configuration (OpenAI, 2025), using native MXFP4 precision, low reasoning effort, and greedy decoding, obtains 19.28 spBLEU and 36.00 chrF++. Separately, our inference-only gpt-oss-120b configuration obtains 23.61 spBLEU on the private test set, below LahjaMT's official constrained and unconstrained scores of 28.30 and 28.54. These configurations are our own experiments, not official shared-task baselines, and are inference-only without task adaptation; the contrast therefore concerns adapted versus unadapted use rather than a like-for-like comparison at equal scale.

| **Method** | **Eval. set** | **$\Delta$ spBLEU** |
|---|---|---|
| Per-sentence oracle [R] | Development | $+5.08$ |
| Metadata-group oracle [R] | Development | $+0.91$ |
| Out-of-fold metadata routing | Development | $-0.17$ |
| Cross-encoder reranking | Development | $-0.71$ |
| LoRA weight interpolation | Development | $-1.03$ |
| gpt-oss-20b inference-only | Development | $-11.65$ |
| Sentence-level MBR | Held-out | $-0.09$ |
| ALLaM-7B post-editing | Held-out | $-0.42$ |
| EG external specialist | Held-out | $-0.19$ |
| YE external specialist | Held-out | $-0.34$ |

Table 4: Diagnostic bounds and unsuccessful alternatives relative to the corresponding routed LahjaMT baseline. [R] denotes reference-dependent selection that cannot be deployed. LoRA interpolation reports its best per-dialect change.

## Appendix D Per-Dialect Effect of Continuation and Rerouting

Held-out spBLEU for the Stage-1 routed system versus the continuation-routed system: EG 29.27 $\rightarrow$ 31.88, JO 31.59 $\rightarrow$ 35.50, LB 28.99 $\rightarrow$ 32.29, LY 17.79 $\rightarrow$ 23.39, MA 21.07 $\rightarrow$ 23.31, MR 15.56 $\rightarrow$ 17.94, OM 28.90 $\rightarrow$ 32.57, PS 31.34 $\rightarrow$ 34.26, SA 32.26 $\rightarrow$ 35.25, SD 21.79 $\rightarrow$ 26.15, SY 37.45 $\rightarrow$ 39.38, TN 32.12 $\rightarrow$ 35.23, YE 22.83 $\rightarrow$ 25.23.

## Appendix E Extended Hyperparameters

**Stage 1:** 3 epochs, LR $2 \times 10^{-4}$, cosine, warm-up 0.03, weight decay 0.01, `adamw_torch`, per-device batch 1 with gradient accumulation 8, max length 2,048, BF16, gradient checkpointing, save/eval every 100 steps, seed 3407, 24,930 optimizer steps. **Stage 2:** identical except LR $2 \times 10^{-5}$ and 7,845 steps over 20,920 examples. **Specialists:** 2 epochs, LR $5 \times 10^{-6}$, `adamw_torch_fused`, seed 42, save every 100 steps. **Generation (everywhere):** `num_beams`=4, `do_sample`=False, `length_penalty`=1.0, `early_stopping`=True, `repetition_penalty`=1.05, `max_new_tokens`=120, left padding and left truncation.